



US012417364B2

(12) **United States Patent**
Gurevich et al.

(10) **Patent No.:** **US 12,417,364 B2**

(45) **Date of Patent:** ***Sep. 16, 2025**

(54) **MINIATURE LONG RANGE IMAGING ENGINE WITH AUTO-FOCUS AND AUTO-ZOOM SYSTEM**

(58) **Field of Classification Search**

None

See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

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(21) Appl. No.: **18/233,609**

Primary Examiner — Kristy A Haupt

(22) Filed: **Aug. 14, 2023**

(57) **ABSTRACT**

(65) **Prior Publication Data**

US 2023/0385576 A1 Nov. 30, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/498,421, filed on Oct. 11, 2021, now Pat. No. 11,790,197.

(51) **Int. Cl.**

G06K 7/10 (2006.01)

G03B 13/36 (2021.01)

(Continued)

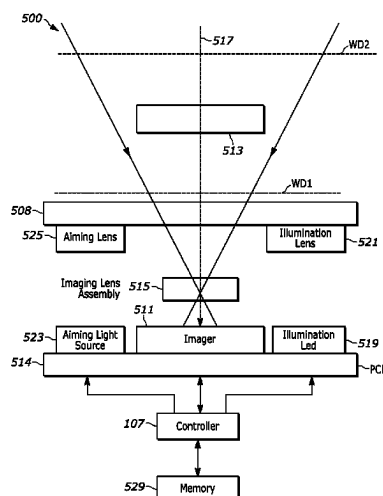
(52) **U.S. Cl.**

CPC **G06K 7/10881** (2013.01); **G03B 13/36** (2013.01); **H04N 23/56** (2023.01);

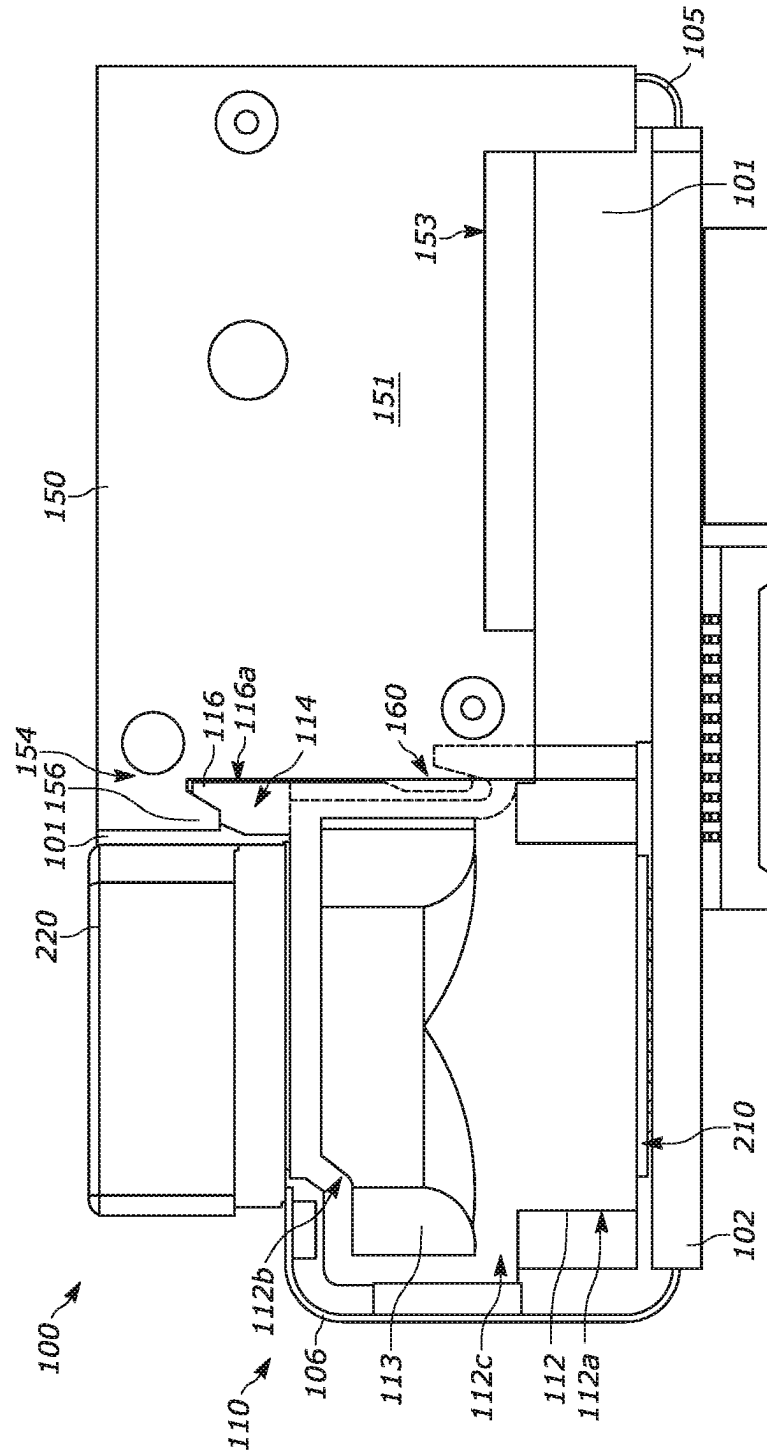
(Continued)

Methods and systems to implement a miniature long range imaging engine with auto-focus, auto-zoom, and auto-illumination are disclosed herein. An example method includes detecting, by a microprocessor, a presence of an aim light pattern within the FOV; determining, by the microprocessor and in response to the detecting, a target distance of an object in the FOV based on a position of the aim light pattern in the FOV, the target distance being a distance from the imaging engine to the object; causing, by the microprocessor, a variable focus optical element to focus on the object based on the target distance; and responsive to making a first determination, by the microprocessor, selecting, based on the target distance, one of a plurality of zoom operation modes.

22 Claims, 12 Drawing Sheets



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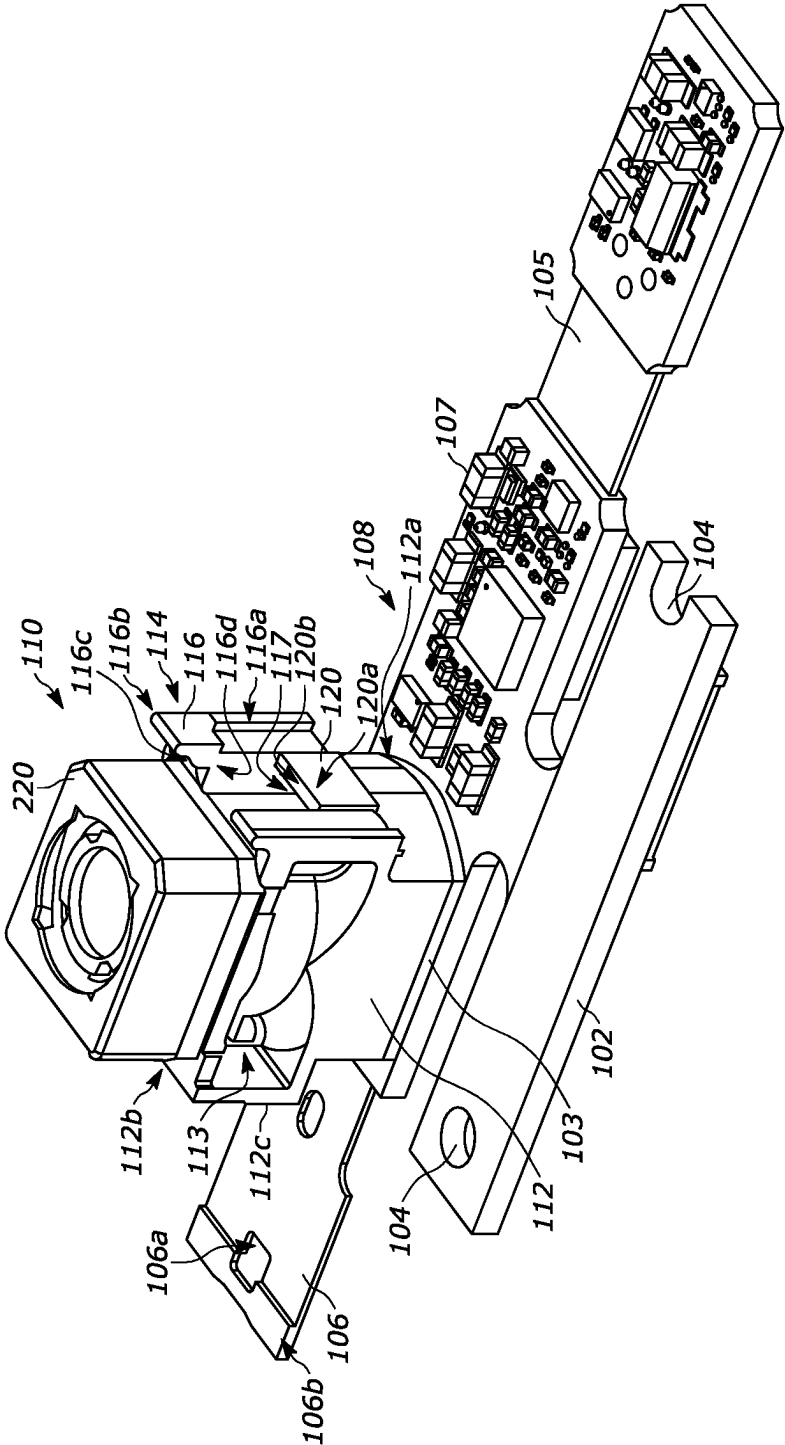
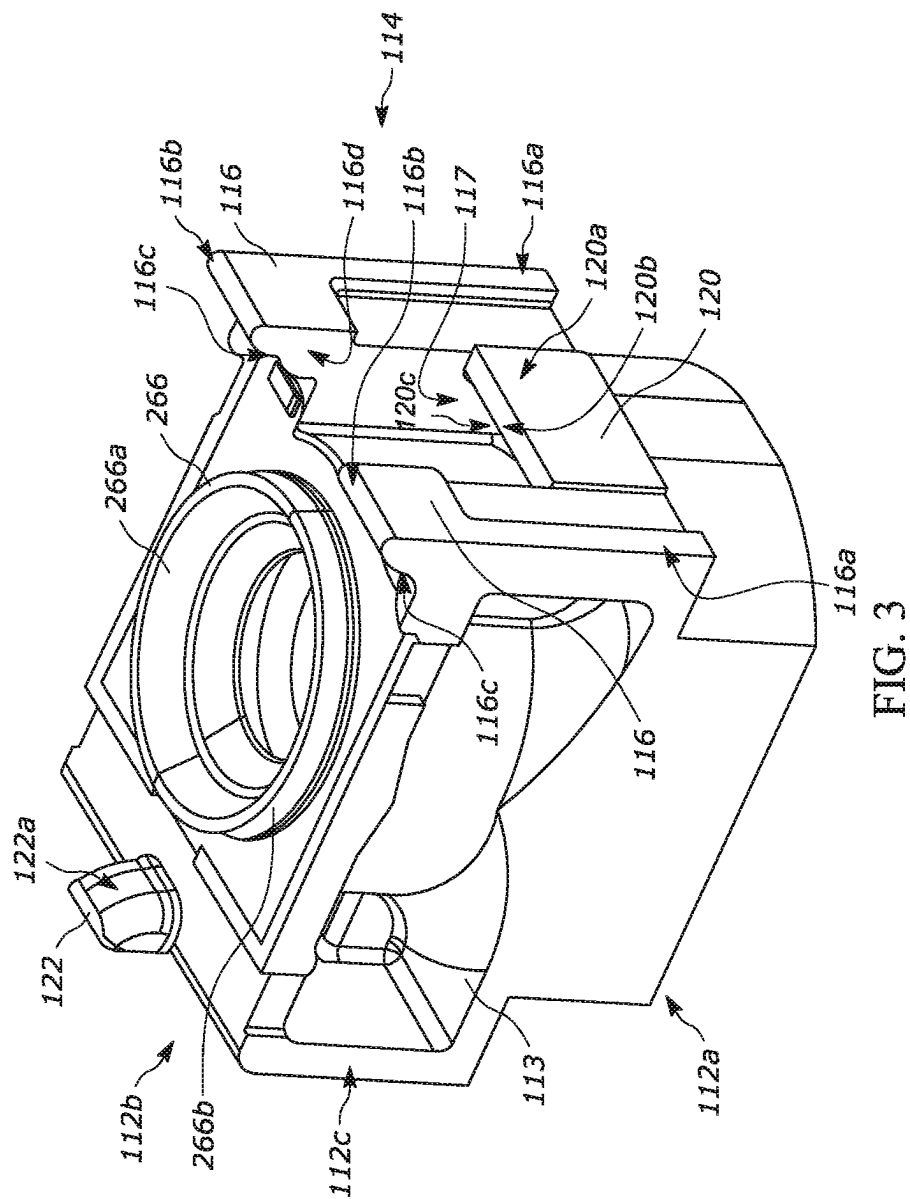


FIG. 2



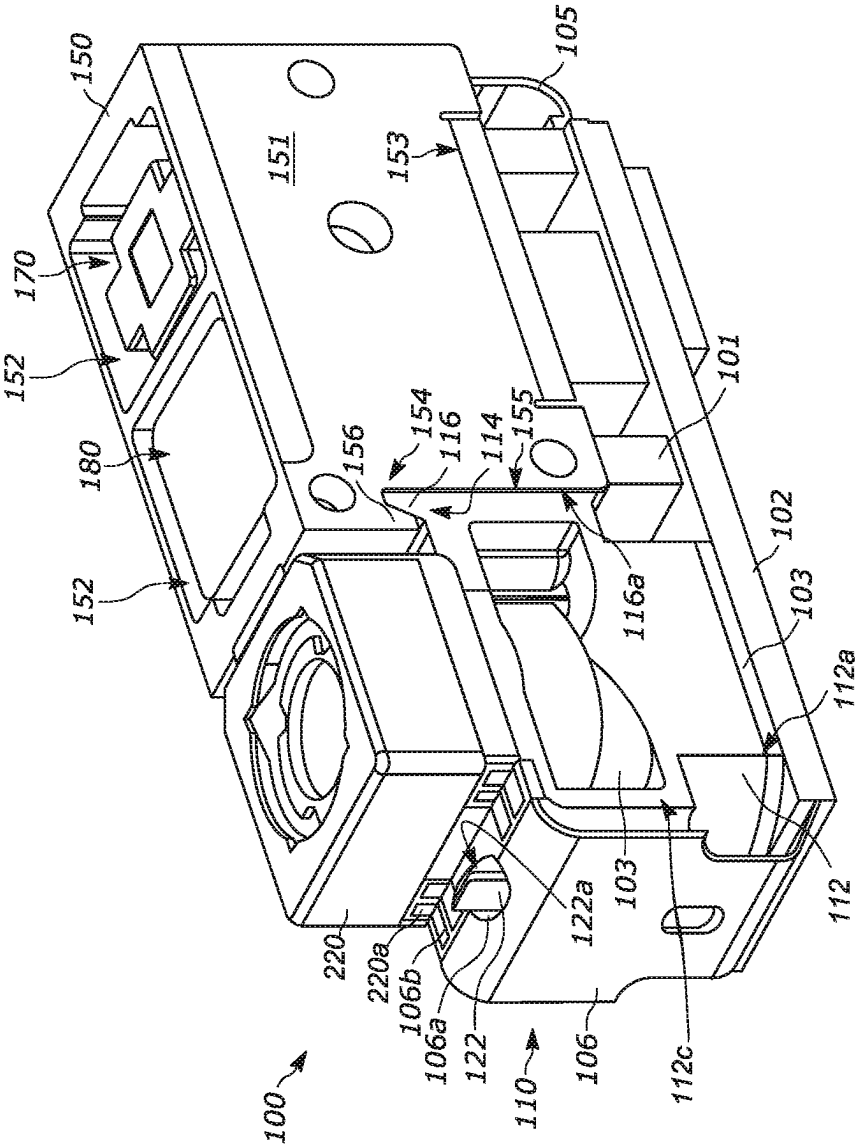


FIG. 4

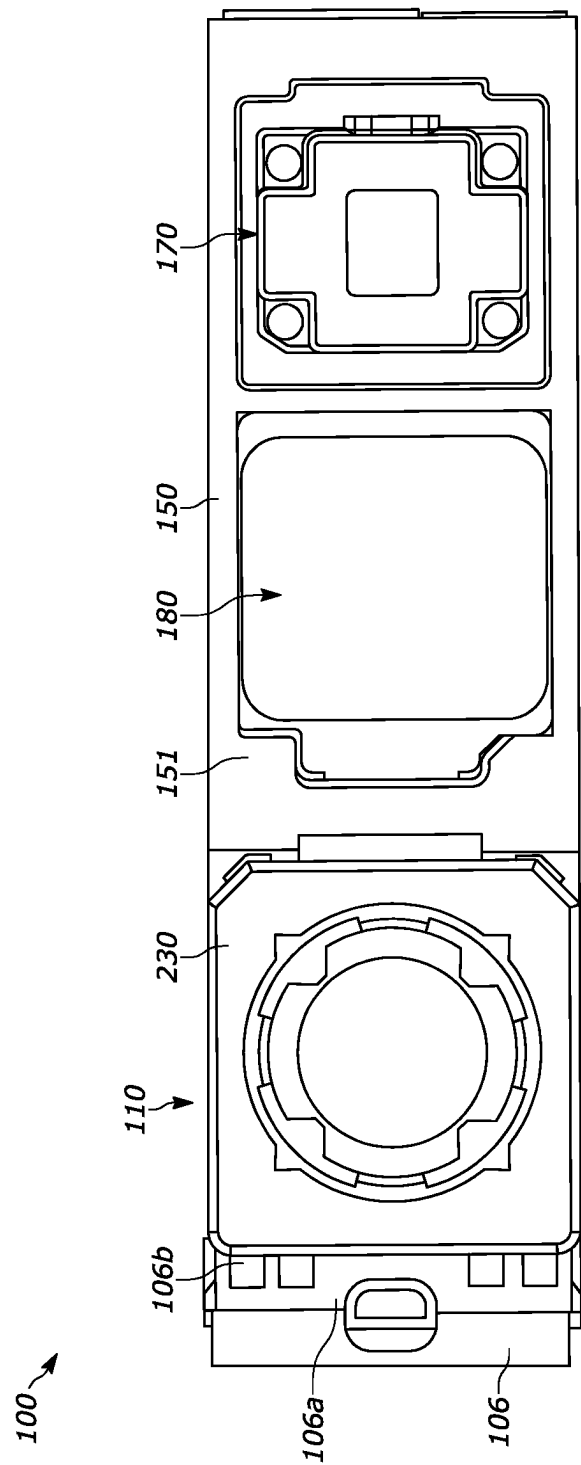


FIG. 5

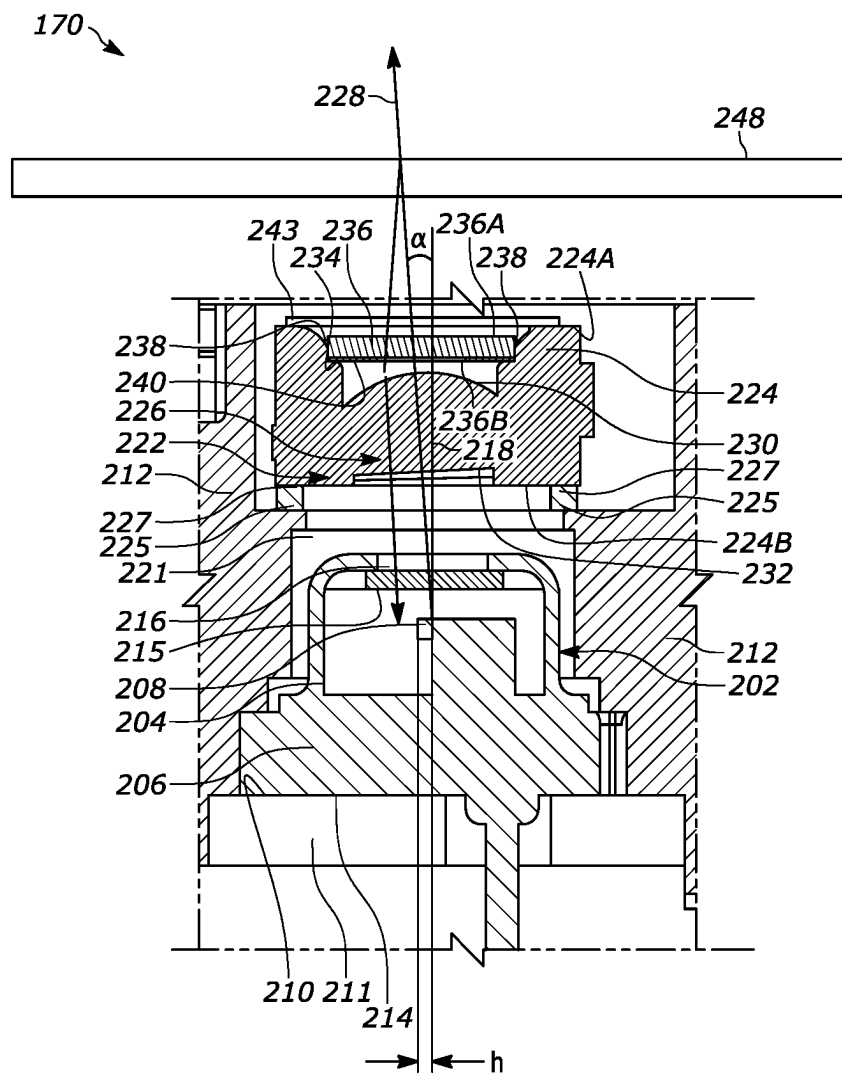


FIG. 6

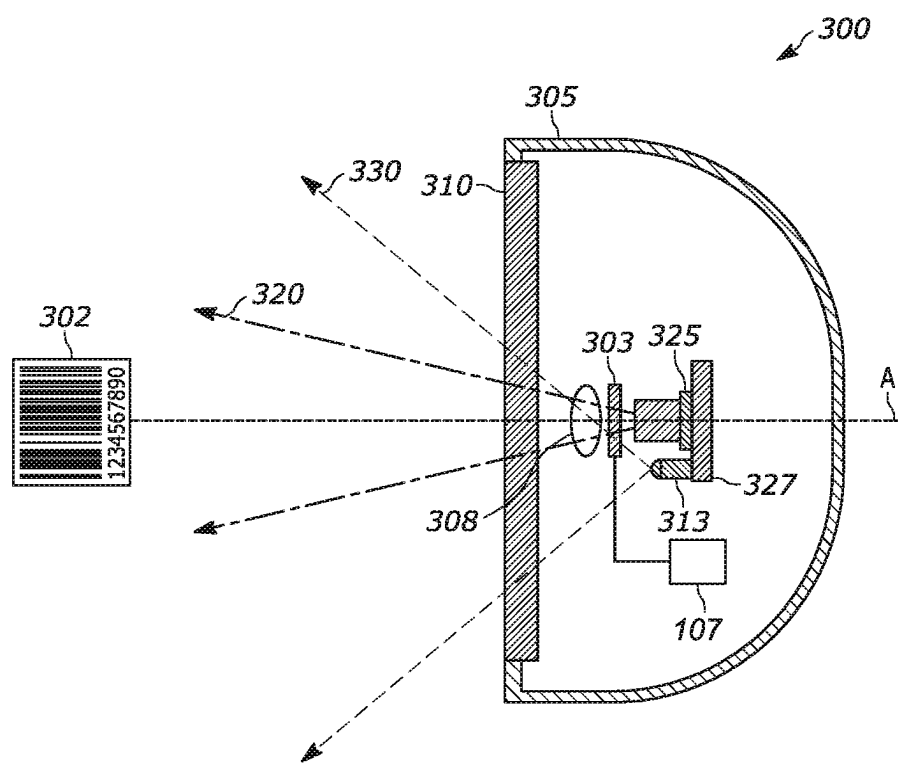


FIG. 7

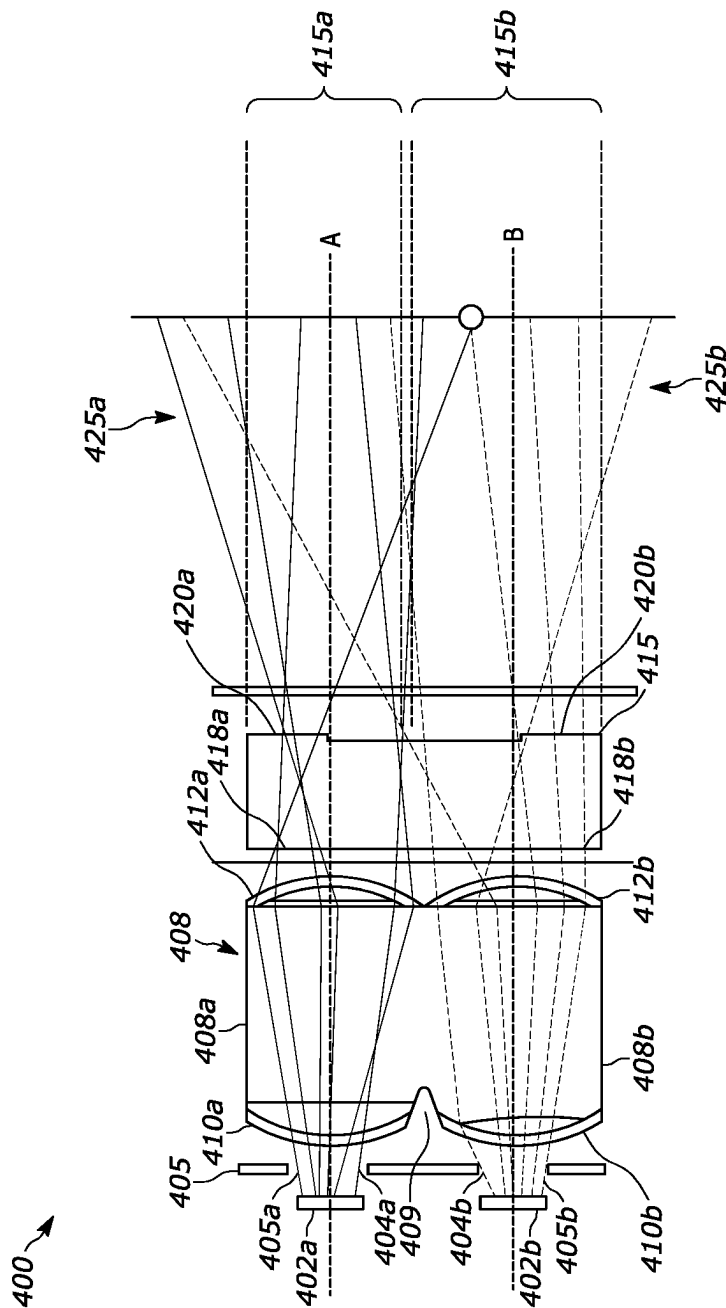


FIG. 8

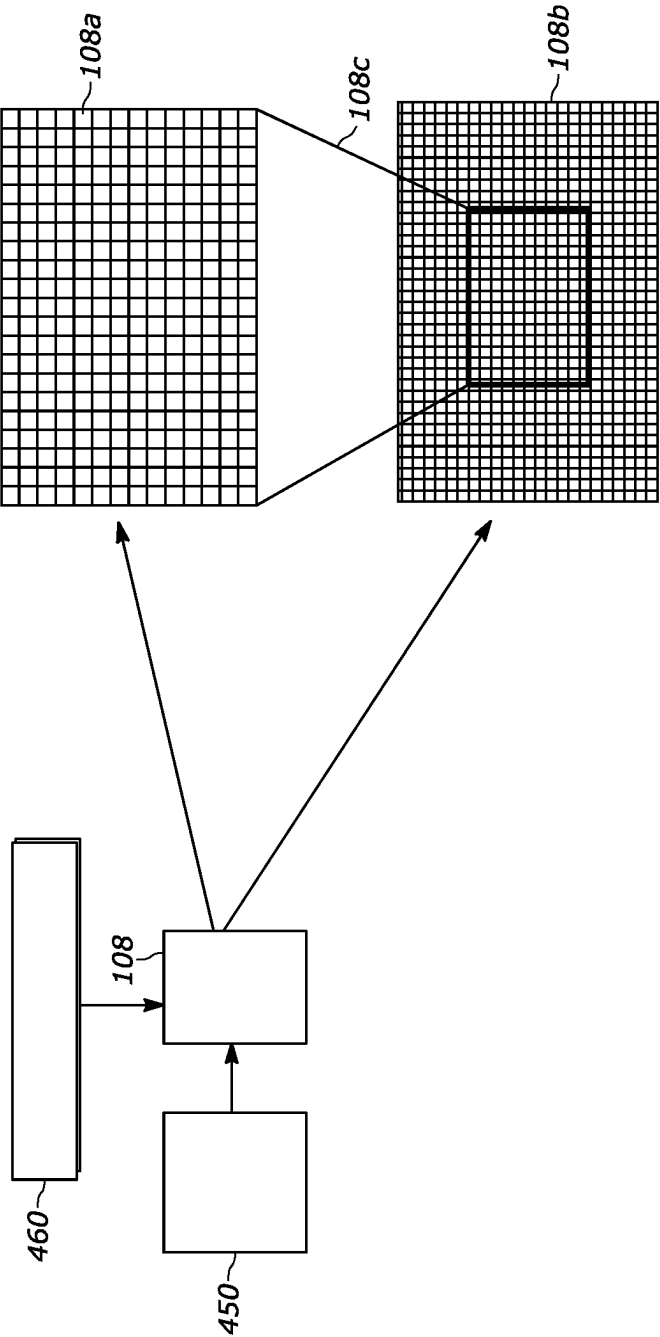


FIG. 9

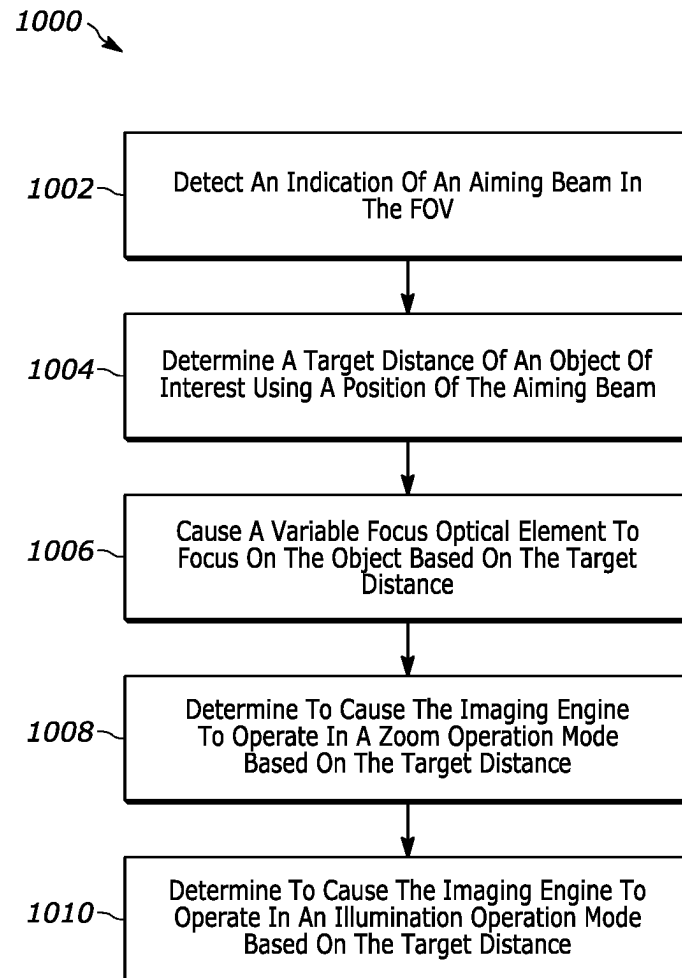


FIG. 10

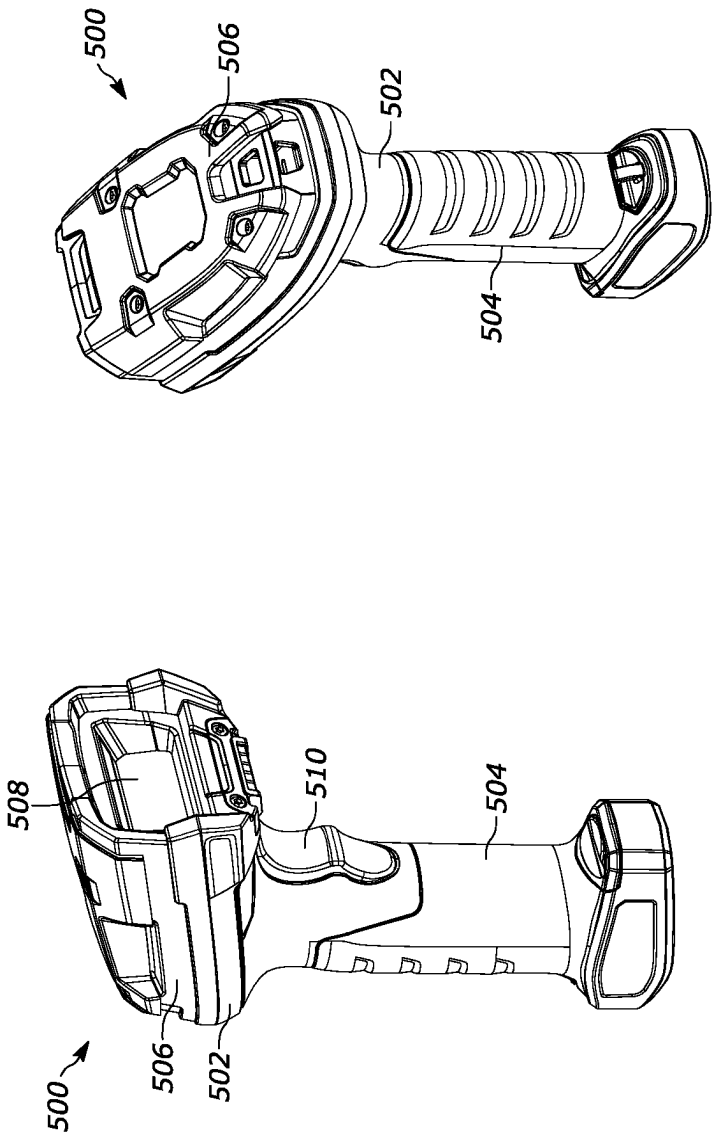


FIG. 11

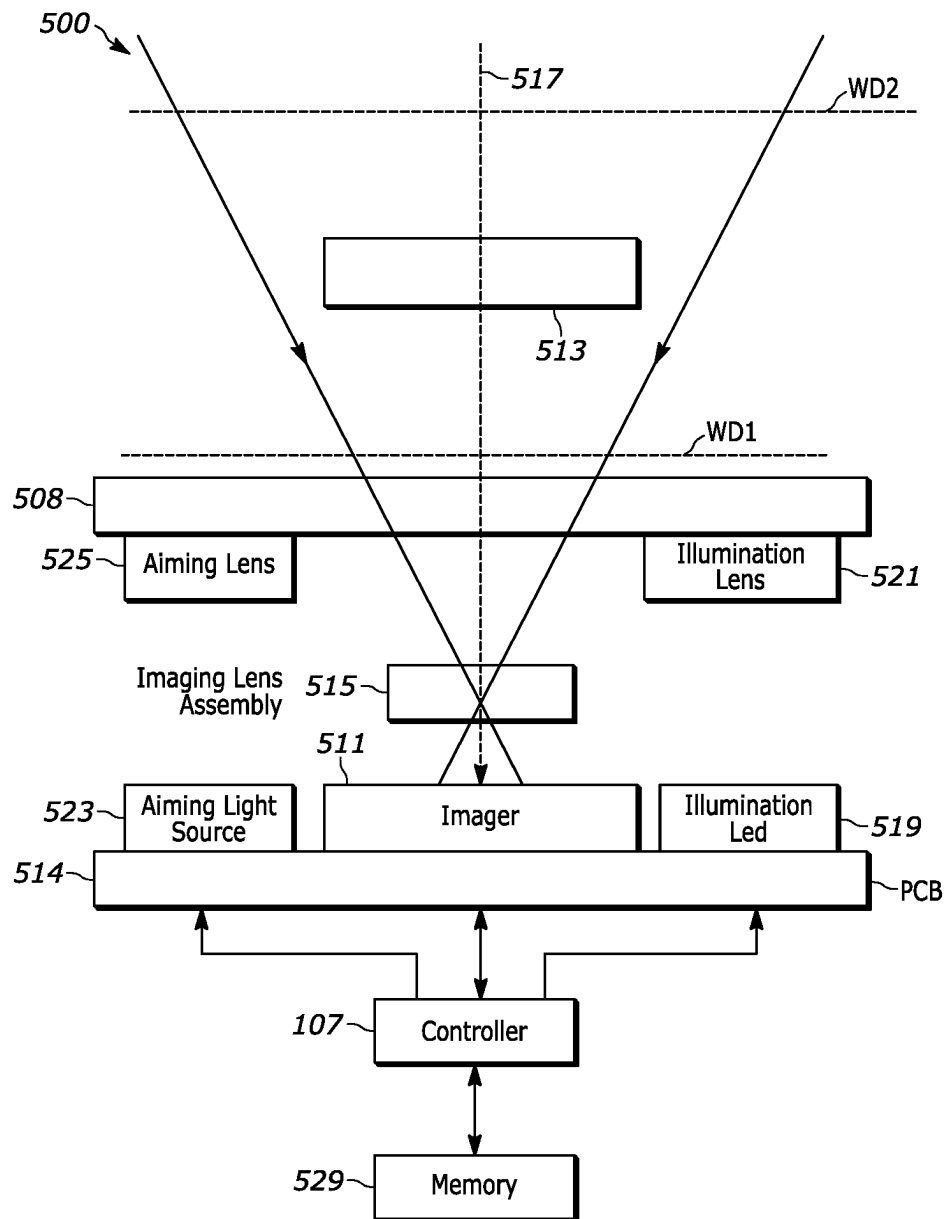


FIG. 12

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MINIATURE LONG RANGE IMAGING ENGINE WITH AUTO-FOCUS AND AUTO-ZOOM SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/498,421, filed on Oct. 11, 2021, and incorporated herein by reference in its entirety.

BACKGROUND

Industrial scanners and/or barcode readers may be used in warehouse environments and/or other similar settings. These scanners may be used to scan barcodes and other objects. In some environments, high powered scanners capable of scanning or resolving barcodes (e.g., 5 millimeter to 100 millimeter wide, Code 128 barcodes) across a wide range of distances, such as from a few inches to tens of feet, or more, may be desirable. Such systems require larger optics (e.g., imaging lens systems greater than approximately 6 millimeters in overall diameter) in order to meet performance requirements, but there remains a compromise between the lens system having a specific size while being constrained by the overall dimensions of the housing and the chassis. Further, compact imaging systems require high precision alignment of optics to prevent optical distortion, which can result in reduced efficiency of scanning rates, or faulty equipment. Moreover, accurately scanning barcodes over a wide range of distances in a various environments requires appropriate focus, illumination, and zoom capabilities. Accordingly, there is a need for improved accessories having improved functionalities.

SUMMARY

In an embodiment, the present invention is a method for range finding and for detecting and imaging objects using an imaging engine having an imaging assembly having a field of view. The method includes detecting, by a microprocessor, a presence of an aim light pattern within the FOV; determining, by the microprocessor and in response to the detecting, a target distance of an object in the FOV based on a position of the aim light pattern in the FOV, the target distance being a distance from the imaging engine to the object; causing, by the microprocessor, a variable focus optical element to focus on the object based on the target distance; responsive to making a first determination, by the microprocessor, selecting, based on the target distance, one of a plurality of zoom operation modes; and, responsive to making a second determination, by the microprocessor, selecting, based on the target distance, one of a plurality of illumination modes.

In a variation of this embodiment, the plurality of zoom operation modes includes at least two of: (i) an image binning mode, (ii) an image cropping mode, and (iii) an image interleaving mode.

In another variation of this embodiment, selecting one of the plurality of zoom operation modes includes: responsive to determining the target distance to be less than a lower threshold value, selecting the image binning mode; responsive to determining the target distance to be greater than an upper threshold value, selecting the image cropping mode; and, responsive to determining the target distance to be between the lower threshold value and the upper threshold value, selecting the image interleaving mode.

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In yet another variation of this embodiment, the lower threshold value is at most 12 inches and the upper threshold value is at least 24 inches.

In still yet another variation of this embodiment, the plurality of illumination operation modes includes at least two of: (i) a power saving mode, (ii) a near illumination mode, and (iii) a far illumination mode.

In another variation of this embodiment, selecting one of the plurality of illumination operation modes includes: responsive to determining the target distance to be less than a lower threshold value, selecting the power saving mode; responsive to determining the target distance to be greater than an upper threshold value, selecting the far illumination mode; and, responsive to determining the target distance to be between the lower threshold value and the upper threshold value, selecting the near illumination mode.

In yet another variation of this embodiment, the lower threshold value is at most 24 inches and the upper threshold value is at least 24 inches.

In still yet another variation of this embodiment, the microprocessor transmits a signal to cause the imaging engine to change to one of the plurality of illumination operation modes after a predetermined delay period elapses after making the determination.

In another variation of this embodiment, the microprocessor determines to change to a different one of the plurality of illumination operation modes during the predetermined delay period and the method further includes changing the signal based on the different one of the plurality of illumination operation modes before transmitting the signal; and resetting the predetermined delay period in response to the updating.

In yet another variation of this embodiment, the variable focus optical element is a ball-bearing motor lens.

In still yet another variation of this embodiment, the object is a barcode, and further includes: cropping a region of interest (ROI) including the barcode and decoding the barcode.

In another variation of this embodiment, the method further includes displaying, to a user, the target distance on a display communicatively coupled to the microprocessor.

In another embodiment, the present invention is an imaging engine for range finding and detecting objects, the imaging engine having an imaging assembly having a field of view (FOV). The imaging engine comprises a variable focus optical element disposed along an optical axis to receive light from an object of interest; an imaging sensor disposed along the optical axis to receive light from the variable focus optical element; a digital zoom module configured to modify an image received from the imaging sensor; an aiming module configured to generate and direct an aim light pattern; an illumination module configured to provide first illumination along a first illumination axis and second illumination along a second illumination axis, the illumination axis not coaxial with the first illumination axis; and a microprocessor and computer-readable media storing machine readable instruction that, when executed, cause the imaging engine to: detect a presence of the aim light pattern in the FOV; in response to the detecting, determine a target distance of the object in the FOV based on a position of the aim light pattern in the FOV, the target distance being a distance from the imaging engine to the object; responsive to making a first determination, select, based on the target distance, one of a plurality of zoom operation modes; and responsive to making a second determination, select, based on the target distance, one of a plurality of illumination operation modes; wherein the variable focus optical ele-

ment, the digital zoom module, the aiming module, and the illumination module are communicatively coupled to the microprocessor.

In a variation of this embodiment, selecting one of the plurality of zoom operation modes includes: responsive to determining the target distance to be less than a lower threshold, selecting an image binning mode; responsive to determining the target distance to be greater than an upper threshold value, selecting an image cropping mode; and responsive to determining the target distance to be between the lower threshold value and the upper threshold value, selecting an image interleaving mode.

In another variation of this embodiment, the digital zoom module is configured to, responsive to selecting the image binning mode, bin pixels of the image using at least one of: 2x2 pixel binning, 3x3 pixel binning, or 4x4 pixel binning.

In yet another variation of this embodiment, the digital zoom module is configured to, responsive to selecting the image cropping mode, crop a portion of the image sized to at least one quarter of the image.

In still yet another variation of this embodiment, the digital zoom module receives the image with a resolution of at least 3 megapixels and zooms on the image with a resolution in a range of 0.5 to 2 megapixels.

In another variation of this embodiment, selecting one of the plurality of illumination modes includes: responsive to determining the target distance to be less than a lower threshold, selecting reduced power mode; responsive to determining the target distance to be greater than an upper threshold value, selecting a far illumination mode; and responsive to determining the target distance to be between the lower threshold value and the upper threshold value, selecting a near illumination mode.

In yet another variation of this embodiment, selecting the zoom operation mode includes: responsive to determining the target distance to be less than a first lower threshold, selecting an image binning mode; responsive to determining the target distance to be greater than a first upper threshold value, selecting an image cropping mode; and responsive to determining the target distance to be between the first lower threshold value and the first upper threshold value, selecting an image interleaving mode; and wherein selecting the illumination operation mode includes: responsive to determining the target distance to be less than a second lower threshold, selecting reduced power mode; responsive to determining the target distance to be greater than a second upper threshold value, selecting a far illumination mode; and responsive to determining the target distance to be between the second lower threshold value and the second upper threshold value, selecting a near illumination mode.

In still yet another variation of this embodiment, the first upper threshold value and the second upper threshold value are equal.

In another variation of this embodiment, the first upper threshold value and the second upper threshold value are at least 40 inches, the first lower threshold value is at most 8 inches, and the second lower threshold value is at most 24 inches.

In yet another variation of this embodiment, the imaging sensor is a rolling shutter sensor configured to operate in at least (i) a first state wherein an obfuscator of the rolling shutter sensor obfuscates a majority of radiation propagating along the optical axis and (ii) a second state wherein the obfuscator of the rolling shutter sensor transmits a majority of radiation propagating along the optical axis.

In still yet another variation of this embodiment, the rolling shutter sensor is communicatively coupled to the

microprocessor and the machine readable instructions, when executed, further cause the imaging engine to transition the rolling shutter sensor between the first state and the second state.

In another variation of this embodiment, the rolling shutter sensor has a pixel size of at most 2.0 micrometers.

In yet another variation of this embodiment, the illumination module includes at least: a first illumination source configured to provide the first illumination; a second illumination source configured to provide the second illumination; a collimator element configured to collimate the first illumination and the second illumination; and a microlens array element configured to receive the first illumination and the second illumination from the collimator element and further to provide a first output illumination field and a second output illumination field.

In still yet another variation of this embodiment, the first illumination source includes a first white LED and the second illumination source includes a second white LED.

In another variation of this embodiment, the first output illumination field corresponds with a first modification of the image and the second output illumination field corresponds with a second modification of the image.

In yet another variation of this embodiment, at least one of the first output illumination field or the second illumination field extends at least 170 inches with no ambient light.

In still yet another variation of this embodiment, the aiming module includes at least a beam source assembly having a beam source for generating the aim light pattern from an exit surface, wherein the exit surface defines a central axis along which an input light is to propagate; and a collimator assembly having a lens group that defines a tilt axis, wherein the tilt axis has a tilt angle relative to the central axis and the lens group is positioned to aim light pattern from the central axis onto the tilt axis.

In another variation of this embodiment, the aiming module generates and directs the aim light pattern in a pulsed laser driving mode.

In yet another variation of this embodiment, the aim light pattern has a wavelength of at least 505 nanometers and at most 535 nanometers.

In still yet another variation of this embodiment, the variable focus optical element is a ball-bearing motor lens.

In another variation of this embodiment, the ball-bearing motor lens has a pupil diameter of at least 2.0 millimeters and a focus range from 3 inches to infinity.

In yet another variation of this embodiment, the object of interest is a barcode and wherein the machine readable instructions, when executed, further cause the imaging engine to decode the barcode.

In still yet another variation of this embodiment, the system further includes a display communicatively coupled to the microprocessor, wherein the machine readable instructions, when executed, further cause the imaging engine to display the distance to a user on the display.

In another variation of this embodiment, the imaging engine further comprises a chassis including a body defining at least one cavity, wherein each of the variable focus optical element, the imaging sensor, the digital zoom module, the aiming module, the illumination module, and the microprocessor and computer-readable media are each at least partially disposed within the at least one cavity.

In yet another variation of this embodiment, the imaging sensor is a single imaging sensor.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout

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the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

FIG. 1 illustrates a front elevation view of an example scanner containing an example imaging assembly for capturing images of an object in accordance with various embodiments;

FIG. 2 illustrates a perspective view of the example imaging assembly of FIG. 1 in accordance with various embodiments;

FIG. 3 illustrates a perspective view of an example lens holder for use with the example imaging assembly of FIGS. 1 and 2 in accordance with various embodiments;

FIG. 4 illustrates a perspective view of an example scanner containing the example imaging assembly of FIGS. 2 and 3 in accordance with various embodiments;

FIG. 5 illustrates a top plan view of the example scanner of FIGS. 1-4 in accordance with various embodiments;

FIG. 6 illustrates a cross-sectional view of an example aiming module incorporated into the imaging assembly of FIGS. 1-5 in accordance with various embodiments;

FIG. 7 illustrates a cross-sectional view of an example imaging system implemented as a rolling shutter lens scanner and incorporated into the scanner of FIGS. 1-5 in accordance with various embodiments;

FIG. 8 illustrates a cross-sectional view of an example illumination module incorporated into the scanner of FIGS. 1-5 in accordance with various embodiments;

FIG. 9 illustrates a block diagram of an example digital zoom module incorporated into the imaging assembly of FIGS. 1-5 in accordance with various embodiments;

FIG. 10 is an example flow diagram of a method for configuring and controlling various modules of the scanner of FIGS. 1-5 in accordance with various embodiments;

FIG. 11 illustrates a perspective front and back view of an optical imaging reader in accordance with various embodiments; and

FIG. 12 illustrates a schematic block diagram of various components of the reader of FIG. 1 in accordance with various embodiments.

Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

Generally speaking, pursuant to these various embodiments, a high-performance autofocus barcode scanner is provided having reduced dimensional requirements, and a broad range of autofocus distances. The scanner also incorporates optical alignment features that provide very high precision alignment of the imaging optics allowing for the use of smaller, more compact, lenses and optical elements. Further still, the scanner incorporates aiming units that generate aiming beams or aiming patterns using a compact,

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low-profile assembly that protects the aiming unit against detrimental back reflections of the aiming beam, back reflections that can otherwise flash blow out compact scanners as well as illumination units that allow for imaging of targets with little to no ambient light. To accurately focus and/or zoom on a target and control the illumination and aiming units, a controller of the scanner operates and adjusts various modules depending on the distance of a target being scanned.

In particular, a miniature imaging engine capable of providing high-resolution image capture for barcode reading and/or range finding over long distances is desired. Existing engines use multiple cameras to achieve varying fields of view (FOVs), requiring a larger and less efficient device. A single imaging sensor system is therefore preferable to existing engines. To achieve a sufficiently small pixel FOV so as to allow for long reading ranges for such a system, an illumination system is required to provide light for imaging. While an increased pupil for lenses in an imaging engine may increase efficiency, such an increase requires auto-focusing capabilities to mitigate resulting decreased depth of focus. As such, an imaging engine with a single imaging sensor, a variable focusing lens, and a controller and/or microprocessor to control zooming, aiming, and focusing functions is described below.

Turning to the figures, an imaging engine device **100** or scan engine for capturing at least one image of an object appearing in an imaging field of view (FOV) is provided. The imaging engine device **100** includes a circuit board **102**, an imaging system **110** operably coupled with the circuit board **102**, and a chassis **150**. Further, the system **100** includes an aiming system or aiming module **170** and an illumination system or illumination module **180**, as well as any number of additional components used to assist with capturing an image or images of an object.

The circuit board **102** may include any number of electrical and/or electro-mechanical components (e.g., capacitors, resistors, transistors, power supplies, etc.) used to communicatively couple and/or control various electrical components of the imaging engine device **100**. For example, the circuit board **102** may include any number of component mounting portions **103**, illustrated in FIG. 2, to receive components (e.g., the imaging system **110**) to operably couple therewith, and may additionally include a board mounting region **104** used to secure the circuit board **102** with the scanner housing (not illustrated). In the example illustrated in FIG. 2, the circuit board **102** further includes a first flex tail connector **105** and a second flex tail connector **106**. As will be discussed, the first flex tail connector **105** is used to communicatively couple components disposed within the chassis **150** with the circuit board **102**, and the second flex tail connector **106** is used to communicatively couple the circuit board **102** with portions of the imaging system **110** and/or with the aiming module **170**.

In particular, the imaging system **110** may be communicatively coupled to a controller **107** of the circuit board **102**. In some implementations, an optical sensor of the imaging system **110** receives light from one or more lenses of the imaging system **110** and, in response, transmits data such as one or more images to or via the controller **107**. The controller **107** may cause the imaging system **110** or a digital zoom module **108** of the imaging system **110** to digitally zoom on some or all of the one or more images. Depending on the implementation, the one or more images may have a resolution of 4 megapixels, and the zoom may produce images with a resolution of 1 megapixel to be analyzed by the controller **107**. One skilled in the art will understand that

megapixels are approximate resolutions of a sensor and cover a range of potential pixel sizes. In some implementations, the controller **107** may cause the digital zoom module **108** of the imaging system **110** to operate in one of multiple zoom operation modes. In some implementations, the digital zoom module **108** may refer to a software module implemented on controller **107** which causes the imaging system **110** to perform particular functions. Depending on the implementation, some operation modes include a binning mode, a cropping mode, and an interleaved mode. The imaging system **110** may operate in the binning mode by binning pixels in the image (e.g., binning pixels in 2x2 pixel squares). Similarly, the imaging system **110** may operate in the cropping mode by cropping an ROI of the image (e.g., a quarter of the image). The imaging system **110** may also operate in the interleaved mode by combining the binning and cropping mode operations. The controller **107** determines the mode as described in more detail with regard to FIG. **10** below.

The imaging system **110** is also operably coupled with the circuit board **102**. The imaging system **110** includes an autofocus system or autofocus module **220** and a rear lens holder **112**, both containing lenses for imaging. In some implementations, the autofocus module **220** includes a variable focus optical element. Depending on the implementation, the lenses for imaging may be or include the variable focus optical element. In a preferred embodiment, the variable focus optical element is a lens operated and/or adjusted by a ball-bearing motor lens or a voice coil motor (VCM) actuator (i.e., a VCM lens). In implementations in which the variable focus optical element is a ball-bearing motor or VCM lens, the ball-bearing motor or VCM lens may have a pupil diameter of at least 2.0 millimeters and a focus range from 3.0 inches extending infinitely (i.e., to optical infinity). In further embodiments, the variable focus optical element may be any lens or optical element with a similar capability to adjust focus, such as a liquid lens, a T-lens, a ball-bearing focusing actuator and any other similar lens known in the art. Depending on the implementation, the controller **107** may control the autofocus module **220** and/or the variable focus optical element.

The autofocus module **220** is positioned adjacent to and/or operably coupled with the rear lens holder **112**. The rear lens holder **112** is in the form of a generally hollow body that defines a lower portion **112a**, an upper portion **112b**, and a sidewall **112c** extending between the lower and upper portions **112a**, **112b**. The rear lens holder **112** may have any number of features such as shapes and/or cutouts **113** such that the sidewall **112c** has a generally uniform thickness despite its particular shape that corresponds to the shape of the lens or lenses disposed therein. These cutouts **113** reduce overall weight of the rear lens holder **112**, and, due to the uniform thickness of the sidewall **112c**, the rear lens holder **112** is easier to manufacture (e.g., mold via an injection molding machine) as compared with lens holders having varying thickness.

In some examples, the rear lens holder **112** is coupled with the circuit board **102** via the component mounting portion **103**. As a non-limiting example, the component mounting portion **103** may be in the form of a pad onto which the lower portion **112a** of the rear lens holder **112** is pressed. The component mounting portion **103** may include an adhesive to assist in securing the rear lens holder **112** to the circuit board **102**. In other examples, the component mounting portion **103** may include any number of electrical intercon-

nects that receive corresponding electrical interconnects disposed or otherwise coupled with the rear lens holder **112**. Other examples are possible.

Referring next to FIG. **3**, the rear lens holder **112** further includes a lens holder mounting portion **114** positioned on an outer periphery of the sidewall **112c**. The lens holder mounting portion **114** includes any number of upper tabs **116** and any number of lower tabs **120**. As illustrated in FIGS. **2** and **3**, each of the upper tabs **116** includes a generally planar facing surface **116a**, a curved upper surface **116b** positioned adjacent to the facing surface **116a**, an angled surface **116c** positioned adjacent to the curved upper surface **116b**, and an inner sidewall **116d** positioned adjacent to the facing surface **116a**, the curved upper surface **116b**, and the angled surface **116c**. In the illustrated example, the respective inner sidewalls **116d** of each of the upper tabs **116** are arranged such that they face each other. The angled surface **116c** is a generally planar surface that forms an angle relative to the facing surface **116a** of approximately 30°. However, other examples of suitable angles are possible.

Each of the upper tabs **116** are separated by a cavity **117** at least partially defined by the inner sidewall **116d**. The cavity **117** is further defined by the lower tab **120**, which includes a generally planar facing surface **120a**, an upper surface **120b** positioned adjacent to the facing surface **120a**, and an angled surface **120c** positioned adjacent to the upper surface **120b**. The angled surface **120c** is a generally planar surface that forms an angle relative to the facing surface **120a** of approximately 30°. However, other examples of suitable angles are possible. Further, while the upper surface **120b** of the lower tab **120** is illustrated as a generally planar surface, in some examples, the upper surface **120b** of the lower tab **120** may be curved. So configured, the cavity **117** is at least partially defined by the inner sidewalls **116d** of the upper tabs **116**, the sidewall **112c**, and the angled surface **120c** of the lower tab **120**. In some examples, the width of the cavity **117** may gradually decrease from the upper portion **112b** to the lower portion **112a**. The lens holder mounting portion **114** also includes a window **266** configured to allow light to a lens or lens group. In some implementations, the window **266** includes at least an outer shell **266a** and an inner shell **266b**. In further implementations, a lens attached to or part of the window **266** is controlled by an actuator such as a VCM actuator or a ball-bearing focusing actuator. In still further implementations, the window **266** may operate in an open state, a closed state, or a partially-open state based on instructions from the controller **107**. Depending on the implementation, the window **266** may be part of a rolling shutter lens barcode reader as described below with regard to FIG. **7**.

The chassis **150** may be constructed from a rigid material such as a metal or metal alloy (e.g., zinc). The chassis **150** includes a body **151** that defines any number of cavities **152** in which components may be partially or fully disposed. For example, the aiming module **170** and/or the illumination module **180** may be at least partially disposed within the cavity **152** of the chassis **150**. The aiming module **170** may include components to generate a pattern or similar visual indication such as an aiming dot to assist with identifying where the imaging system **110** is aiming. In some examples, the aiming module **170** may include laser and/or light emitting diode ("LED") based illumination sources. The illumination module **180** assists with illuminating the desired target for the imaging system **110** to accurately capture the desired image. The illumination module **180** may include an LED or an arrangement of LEDs, lenses, and the

like. The aiming module **170** and the illumination module **180** are described in more detail with regard to FIGS. **6** and **8** below.

The body **151** of the chassis **150** may include a recessed portion **153** that is adapted to receive a portion of the first flex tail connector **105** (e.g., a sub-board or an interconnect member). The chassis **150** further includes a chassis mounting portion **154** disposed or positioned on an outer periphery of the body **151** of the cavity **150**. The chassis mounting portion **154** further includes any number of upper hooks **156** and any number of lower hooks **160**.

With reference to FIG. **4**, the second flex tail connector **106** includes a mounting opening **106a** and a number of interconnects **106b**. The rear lens holder **112** includes a flex securing tab **122** that protrudes upwardly from the rear lens holder **112**. The flex securing tab **122** includes an angled engaging surface **122a** which is angled in a direction towards the autofocus module **220**. When electrically coupling the autofocus module **220** with the circuit board, the second flex tail connector **106** is urged upward, and the mounting opening **106a** is aligned with the flex securing tab **122**. Because the engaging surface **122a** of the flex securing tab **122** is angled towards the autofocus module **220**, the interconnects **106b** are moved or positioned against corresponding interconnects **220a** positioned on the autofocus module **220**, thereby communicatively coupling the autofocus module **220** with the circuit board **102**. In some examples, the flex securing tab **122** may include a notch or other feature used to retain the second flex tail connector **106**.

So configured, and as illustrated in FIGS. **4** and **5**, the imaging system **110** described herein may occupy an entire available height between the opposing large flat mounting surfaces of the chassis **150** as compared with being constrained by the body **151** of the chassis **150**. Further, instead of the chassis **150** being mounted directly to the circuit board **102**, the imaging system **110** is mounted to the circuit board **102** while the chassis **150** is coupled with the imaging system **110**. Advantageously, such an arrangement isolates heat of the aiming module **170** and the illumination module **180** disposed within the chassis **150** from the optical sensor mounted on the circuit board **102**, while also providing for additional optical path length for the imaging system **110**.

FIG. **6** illustrates an example implementation of the aiming module **170**. The aiming module **170** is configured to generate an aim light pattern to serve as a visual guide for users during operation of the imaging engine device **100**, in particular for accurate positioning of the imaging system **110** and the illumination module **180**. In some implementations, the aim light pattern is an aiming beam. Whereas conventional aiming assemblies are able to generate bright, central aiming dots or patterns, they typically do so on an axis offset from that of the illumination module field of view and imaging system field of view. And for configurations where aiming assemblies are designed to tilt the axis of an aiming beam, the configurations are too large, requiring wedges or similar optics, to be compatible with integrated scanner assemblies, such as those described herein. In contrast, in various examples, aiming modules provide small stack height designs capable of generating aiming dot or aim light pattern axis tilt without increasing overall height of an integrated scanner assembly.

In the example of FIG. **6**, aiming module **170** includes a beam source assembly **202** which has a frame **204** having a mounting plate **206** onto which a beam source **208** is positioned. The beam source **208** may be a laser or LED based beam source. In some examples, the beam source **208**

is a vertical emitting beam source, such as a vertical cavity surface emitting laser. In some examples, the beam source **208** is a side emitting beam source or edge emitting beam source. Depending on the implementation, the aiming module **170** may direct the beam source **208** to generate the aim light pattern and/or beam in a pulsed laser driving mode with a set or variable frequency. In some implementations, the pulsed laser driver mode pulses the laser at a frequency of 20 Hz, 40 Hz, 60 Hz, 80 Hz, or 100 Hz.

In further implementations, the aim light pattern may be red (e.g., the aim light pattern has a wavelength of 630 nanometers to 670 nanometers) or green (e.g., the aim light pattern has a wavelength of 505 to 535 nanometers). Depending on the implementation, the aim light pattern may be limited to an average power of less than or equal to 1 milliwatt but is visible in sunlight at a distance of at least 40 inches.

The frame **204** may be an integrated piece having a mounting surface **210** mountable to a mounting surface **214** of a mounting plate **211**, which may be formed with or attached to a chassis **212** to serve as a chassis mounting portion. In other examples, the frame **204** may be mounted (e.g., glued onto or pressed into) directly on the chassis **212**, without a mounting plate **211**. For example, walls of the lower cavity **221** may be sized to receive the mounting plate **206** of the frame **204** and fixedly retain the later in place. In some examples, the mounting plate **211** and/or the mounting plate **206** may provide a heat dissipation function for the laser **208**.

The frame **204** includes a transparent window **215** environmentally sealing the laser **208** and positioned adjacent an opening **216** that functions as an aperture through which the generated aim light pattern is provided along a beam axis **218**. The frame **204** sits within a lower cavity **221** of the chassis **212**. In some examples, the lower cavity **221** may be environmentally sealed using a transparent window at an upper end (not shown). The chassis **212** further includes an outer cavity **223** having chassis mounting portions (surfaces) **225** onto which the collimator assembly **222** may be placed during assembly and held in place by an adhesive, such as a UV curable adhesive **227** surrounding a lower outer edge of the assembly **222**. Further a transparent window **248** may be mounted to an exit end of the chassis **212**, above the collimator assembly, for transmission of the aiming pattern along a tilt axis as described below.

The collimator assembly **222** is a low profile assembly having a body **224** that has an outer surface **224A** and an inner surface **224B** parallel thereto. The collimator assembly further includes a lens group **226** that is positioned between the outer surface **224A** and the inner surface **224B**. More particularly, the lens group **226** defines a tilt axis **228**. In the illustrated example, that tilt axis **228** forms an acute angle relative the parallel outer and inner surfaces **224A**, **224B**. Further, the tilt axis **228** defines a tilt angle, α , relative to the beam axis **218**, which may also be considered a central axis. Further still, the lens group **226** is positioned relative to the beam source **208** such that aim light pattern and/or beam, incident along the beam axis **218**, is deflected onto the tilt axis **228** by the lens group **226**. In various examples, the tilt angle, α , is confined by the expression $\alpha > 0.5 \cdot \arctan(h/F)$, where F is a focal length of the lens group **226** and h is a clearance height of the beam source **208**, so as to prevent back reflection of the aim light pattern and/or beam from an exit window back on to the beam source **208**.

In various examples, the lens group **226** includes a first lens **230** at an exit end and a second lens **232** at an entrance end. Both of the lens **230** and **232** may be tilted, meaning

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having a central shared axis that is tilted related to the beam axis **218**. In some examples, one or both of first lens **230** and second lens **232** is a semispherical lens, meaning a lens whose surface profile has at least a portion thereof formed of a sphere or cylinder. In some examples, one or both of the first lens **230** and the second lens **232** is an aspheric lens, meaning a lens whose surface profiles are not portions of a sphere or cylinder. In some examples, the lens group **226** may be formed of a double convex lens having a central axis tilted to be parallel with the axis **228**. In some examples, the lens group **226** may be formed of a lens having symmetric aspheric surfaces, where the first lens **230** and the second lens **232** have aspheric curvatures. In other examples, the second lens **232** is implemented as a generally planar surface instead, e.g., a tilted planar face. In various examples, the lens group **226** is integrally formed from the body **224** such that it is a continuous piece.

As shown in FIG. 6, in various examples, the optical element **236** is a diffractive optical element and may have a planar outer surface **236A** and a diffractive element at an inner surface **236B**, the latter being positioned to receive the input beam from the lens group **226**. The sealing adhesive **238**, therefore, not only retains the optical element **236** in the recess **234**, but further provides environmental sealing that prevents contamination of the diffractive element located at the inner surface **236B**. In other examples, the optical element **236** may be a refractive optical element or a combination of diffractive and refractive elements. The optical element **236** is configured to convert the input beam into an aiming pattern that appears at a focal distance of the lens group **226**. This configuration provides a compact design while still allowing for converting an input beam into a complex beam or aim light pattern at the far field, where such patterns can be geometric shapes, such as squares, rectangles, circles, etc. But also more complex shapes such as logos, text, pictures, etc. The optical element **236**, as part of the lens group **226**, may further cooperate with the aspheric surface **230** to collimate the incident input beam along with tilting the propagation axis of the beam. An optical element retainer **243**, also termed an eye-safety retainer, is affixed to the upper surface **224A**, e.g., using an adhesive, to provide further protection against dislodging of the optical element **236** from the aiming module **170**. The retainer **243** may include an aperture positioned and sized to allow transmission of the deflected aiming beam or aim light pattern, where the remainder of the retainer **243** may be non-transparent. The retainer **243**, for example, may be formed with a black material or other opaque medium or even a partially transmitting medium such as a diffuser. The retainer **243**, therefore, allows the deflected aim light pattern through its aperture but otherwise blocks or minimizes back reflections and stray light from entering the lower cavity **221** and impinging on the laser **208**. As such, the retainer **243** prevents stray light from causing spurious reflections and brightness flares. The retainer **243** additionally prevents the optical element **236** from being in the path of the deflected aiming beam or aim light pattern, in the event the adhesive **238** fails. In some examples, the optical element retainer **243** further includes sensing electrodes positioned thereon to allow an external circuit to determine if the window itself has been dislodged (e.g., in response to a received signal, a change in measured impedance, or other electrical detection) providing a warning signal to an operator.

In various examples, a beam forming aperture **240** is placed first in the recess **234** to provide shielding of the lens group **222** against extraneous off axis back scattering light,

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ambient light, or other illumination. The optical element **236** may then be placed on top of that aperture **240**.

Referring to FIG. 7, an example rolling shutter sensor system **300** is shown that uses an obfuscative element or obfuscator **303** to operate as a rolling shutter sensor system **300** for imaging an object of interest **302**. The obfuscator **303** is disposed inside of a housing **305** along an optical path **A** of field of view **320** of an imaging sensor **325**. Depending on the implementation, the obfuscator **303** may be a physical external optical or mechanical shutter. In such implementations, the obfuscator **303** enhances the performance of the rolling shutter sensor system **300**. In further implementations, the obfuscator **303** is part of the imaging sensor **325** and allows the rolling shutter sensor system **300** to function as a rolling shutter sensor system **300**. In other implementations, the obfuscator **303** is not present and the rolling shutter sensor system **300** operates using electronic means instead (e.g., by filtering out light received by the imaging sensor).

In some implementations, the housing **305** is contained within and/or part of rear lens holder **112**. In further implementations, the housing is a separate part of the chassis **150**. Similarly, depending on the implementation, the imaging sensor **325** is or includes the sensor of the imaging assembly **110**. A controller **107** controls a state of the obfuscator between an obfuscative state and a transmissive state. The obfuscative state is an optical state in which the obfuscator obfuscates or obscures a majority of radiation to the imaging sensor **325** along the optical axis **A**, and the transmissive state is an optical state in which the obfuscator transmits a majority of radiation to the imaging sensor **325** along the optical axis **A**. In implementations in which the imaging sensor **325** is a rolling shutter sensor, the obfuscator may move to allow some radiation to the imaging sensor **325** in smaller subsets—i.e., the imaging sensor **325** receives light in a rolling pattern. In further implementations, the imaging sensor **325** may be disposed and/or configured to receive light in a rolling pattern without the obfuscator **303**.

The imaging sensor **325** is mounted on an imaging circuit board such as circuit board **327**, which may provide power to the imaging sensor **325**, control of operation of the sensor **325**, on/off board communications of data to and from the imaging sensor **325**, among other operations and purposes. In some implementations, the imaging circuit board **327** is part of or is circuit board **102**. The imaging sensor **325** may be a CMOS device, or other imaging sensor capable of functionality as a rolling shutter sensor. In some implementations, the imaging sensor **325** is the optical sensor on the circuit board **102** and/or is part of imaging system **110**. The imaging sensor **325** may have a fixed exposure time, or the exposure time and rolling shutter functionality may be tuned to change the exposure time based on an object of interest, a distance of the object of interest, an illumination of the object of interest, etc. For example, in some implementations, the imaging sensor **325** exposure time and rolling shutter functionality may be tuned to operate in varying modes depending on the speed of a target (e.g., low-speed, high-speed, and very high-speed modes). In a particular preferred embodiment, the imaging sensor has a pixel size of at most 2.0 micrometers.

A lens **308** is disposed along the optical path **A** to focus images received by the rolling shutter sensor system **300** onto an imaging plane at the imaging sensor **325**. A window **310** is disposed along the optical axis **A** to provide a transmissive surface for optical radiation to pass along the optical axis into the housing **305**. In some implementations, the window **310** is the window **266** discussed with regard to

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FIG. 3 above. The window 310 acts as an aperture and may be useful for preventing stray light and optical noise from entering the housing 305. Further, the window may be a material or have coatings to perform as a filter to reduce noise or to select wavelengths of light for imaging at the imaging sensor 325. Each of the window 310, lens 308, and obfuscator 303 are disposed to image the object of interest 302 onto the imaging sensor 325.

While not illustrated, a person of ordinary skill in the art would recognize that additional or fewer optical elements may be implemented along the optical axis for imaging of the object of interest. For example, one or more additional lenses, wavelength filters, spatial filters, polarizers, beam splitters, mirrors, waveplates, apertures, or other optical elements may be employed for imaging of the object of interest 302. In a configuration, the object of interest 302 includes one or more indicia indicative of information about the object of interest, the indicia being one or more of a 1D or 2D barcode, QR code, dynamic QR code, UPC code, serial number, alphanumeric, a graphic, or another indicia.

The obfuscator 303 may be a transfective mirror, positioned within the housing 305 along the optical path A. As a transfective mirror, the obfuscator 303 can be switched between a transmissive state, in which a majority of light is allowed to pass through the transfective mirror, and a reflective state, in which a majority of light is reflected off of the transfective mirror. For example, the obfuscator 303 may switch states in response to an electrical control signal received from the controller 107. With the transfective mirror in the reflective state, the transfective mirror reflects at least a first portion of radiation in the field-of-view 320 of the imaging sensor 325. In the transmissive state, the transfective mirror allows for optical radiation within the field-of-view 320 to pass through the transfective mirror 355 along the optical path A to the imaging sensor 325. Optionally, the transfective mirror could also be switched to a partially reflective state, in which the transfective mirror would both reflect a portion of light, and transmit a portion of light. Such an example may be useful in a system that images the object of interest 302 while targeting radiation is provided to the barcode or object of interest. For example, the rolling shutter sensor system 300 may further include a target radiation source 313 that provides radiation 330 to the object of interest 302 for a user of the barcode reader to reference when positioning the object of interest for scanning, or when position the rolling shutter sensor system 300 in the case of a handheld barcode reader.

While described above as a transfective device, the obfuscator 303 does not need to reflect optical radiation. In the obfuscative state, the obfuscator 303 may absorb the radiation, or otherwise obscure the optical radiation to prevent the radiation from reaching the imaging sensor 325, while the obfuscator 303 passes radiation to the imaging sensor 325 when in the transmissive state. In configurations, the obfuscator 303 may include one or more of a transfective mirror, a different transfective element, an electrochromic device, a polymer-dispersed liquid crystal film, or another electrically controllable shutter element (e.g., an external shutter) capable of transitioning between states at a time scale operational for a rolling shutter sensor system 300.

Further, while FIG. 7 illustrates an imaging engine 100 with a rolling shutter sensor system 300 using an example rolling-shutter sensor, imaging engine 100 and/or rolling shutter sensor system 300 may include any similar such sensor. For example, a global shutter sensor—i.e., a sensor in which all pixels of an array are exposed simultaneously

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rather than individual subsets of pixels being exposed at different times—may be used. Similarly, a lens or sensor system capable of operating in both a rolling-shutter or global shutter mode may also be used.

Next, FIG. 8 illustrates a cross-sectional side view of a ray trace of an embodiment of an optical assembly 400 of a dual illumination module 180. The optical assembly 400 includes a first illumination source 402a and a second illumination source 402b. The first illumination source 402a is disposed along a first optical axis, A, to provide first illumination 404a along the first optical axis, A. The second illumination source 402b is disposed along a second optical axis, B, configured to provide second illumination 404b along the second optical axis, B. In some implementations, the first and second illumination sources 402a and 402b may include one or more LEDs, laser diodes, lasers, black body radiation sources, or other such illumination sources. In embodiments, the first and second illumination 404a and 404b may include one or more of infrared radiation, near-infrared radiation, visible light, optical radiation, ultraviolet radiation, or another type of radiation for illumination of a target for imaging of the target.

The first and second illumination sources 402a and 402b may be square light sources and center points of the first and second illumination sources 402a and 402b may be disposed between 1 and 5 millimeters apart, between 5 and 10 millimeters apart, less than 10 millimeters apart, or greater than 1 centimeter apart. Further, the first and second illumination sources 402a and 402b may be 1 millimeter by 1 millimeter squares, 2 millimeters by 2 millimeters squares, 5 millimeters by 5 millimeters squares, or larger than 5 millimeters by 5 millimeters squares. In a particular implementation, the first and second illumination sources 402a and 402b are 1 millimeter by 1 millimeter square white LED lights. Depending on the implementation, the first and second illumination sources 402a and 402b may have sufficient luminance to allow for barcode reading at a distance of up to 170 inches in ambient darkness, and/or greater distances in at least low levels of ambient light (e.g., 5-10 foot-candles). Similarly, depending on the implementation, the first and second illumination sources 402a and 402b output illumination fields 425a and 425b in rectangular shapes, cone shapes, or any other suitable design.

The first and second illumination sources 402a and 402b may also be circular, rectangular, or another geometric shape. The optical assembly includes an aperture element 405 having a first aperture 405a and a second aperture 405b. The first illumination 404a propagates along the first optical axis A through the first aperture 405a, and the second illumination 404b propagates along the second optical axis B through the second aperture 405b. Depending on the implementation, the optical axis A may or may not be the same optical axis A referred to in FIG. 7. The first and second apertures 405a and 405b may be independent apertures, or they may be two apertures of a same larger aperture element, such as two holes or openings in a single material with the two holes being independent and spatially separated by some distance. Further, the first and second apertures 405a and 405b may be a same large aperture that transmits both the first and second illumination 404a and 404b.

A collimator element 408 is disposed along the first and second optical axes A and B to collimate the first and second illumination 404a and 404b. The collimator element 408 has a first collimator 408a and a second collimator 408b. The first collimator 408a has a first collimator entry surface 410a configured to receive the first illumination 404a from the first aperture 405a, and the second collimator 408b has a

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second collimator entry surface **410b** configured to receive the second illumination **404b** from the second aperture **405b**. The first and second collimator entry surfaces **410a** and **410b** may be separated by a separator element **409** that prevents at least some of the first illumination **404a** from entering the second collimator **408b**, and further prevents at least some of the second illumination **404b** from entering the first collimator **408a**. The separator element **409** may include a wedge or wall of air, metal, plastic, glass, or another material. The first collimator **408a** has a first collimator exit surface **412a** disposed along the first optical axis A to provide collimated first illumination **404a** to a microlens array element **415**. The second collimator **408b** has a second collimator exit surface **412b** disposed along the second optical axis B to provide collimated second illumination **404b** to the microlens array element **415**.

The microlens array element **415** is disposed along the first and second optical axes, A and B respectively, to receive the collimated first and second illumination **404a** and **404b** from the collimator element **408**. The microlens array element **415** has a first microlens array **415a** and a second microlens array **415b**. The first microlens array **415a** has a first microlens entry surface **418a** disposed along the first optical axis A to receive the first illumination **404a**. The first microlens array **415a** also has a first microlens exit surface **420a** to provide the first illumination **404a** as a first output illumination field **425a**, illustrated by solid lines in FIG. 8, to a target for imaging of the target. The second microlens array **415b** has a second microlens entry surface **418b** disposed along the second optical axis B to receive the second illumination **404b**. The second microlens array **415b** also has a second microlens exit surface **420b** to provide the second illumination **404b** as a second output illumination field **425b**, illustrated as broken lines in FIG. 8, to a target for imaging of the target. In some implementations, each of the first output illumination field **425a** and second output illumination field **425b** correspond with an operation mode of illumination module **180** as described with regard to FIG. 10 below. For example, the first output illumination field **425a** may illuminate a near FOV portion of the overall FOV while the second output illumination field **425b** illuminates a far FOV portion.

Each of the first and second microlens arrays **415a** and **415b** may each independently spread input radiation or stretch an input radiation field to provide an output illumination field with one or more dimensions having a wider field angle than input collimated illumination. The microlens array element **415** may be a plastic material such as Zeonex, Acrylic Polycarbonate, K26R, E48R, or another such material. In some implementations, the microlens array element **415** may be a glass material or other optical material able to transmit light. Further, the distance between either of the first and/or second illumination sources **402a** and **402b** to the second surface of either of the first and/or second microlens exit surfaces **420a** and **420b** may be 5 millimeters, 7 millimeters, 10 millimeters, 12 millimeters, less than millimeters, less than 10 millimeters, or less than 8 millimeters to provide a compact form factor for the optical assembly **400**.

FIG. 9 illustrates an example schematic block diagram of the digital zoom module **108**. As is noted above, depending on the implementation, the digital zoom module **108** may be implemented in a separate piece of hardware than controller **107** or may be a software module implemented on the controller **107**. The digital zoom module **108** receives an image **450** taken by the imaging system **110** or otherwise received by the controller **107**. The image may have a

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resolution of 1 megapixel, 2 megapixels, 4 megapixels, 6 megapixels, 8 megapixels, or any other suitable resolution for imaging. In a particular preferred embodiment, the image has a resolution of at least 3 megapixels. The digital zoom module **108** also receives a target distance **460**. Depending on the implementation, the controller **107** may calculate or otherwise determine the target distance **460** using methods outlined above and subsequently analyze the target distance **460** using the digital zoom module **108**. In other implementations, the digital zoom module **108** may receive the target distance **460** from the controller **107** or from another piece of hardware in the digital imaging engine **100**. Depending on the target distance **460**, the digital zoom module begins operation in an operation mode. In some implementations, the digital zoom module **108** may operate in multiple operation modes and the controller **107** determines which operation mode the digital zoom module **108** is to operate in. In further implementations, the digital zoom module **108** may operate in one of three operation modes: a binning mode **108a**, a cropping mode **108b**, and an interleaving mode **108c**. In the binning mode **108a**, the digital zoom module **108** takes individual pixels and combines the pixels into larger pixels (i.e., super-pixels). In some implementations, the digital zoom module **108** may bin the pixels into super-pixels with a size of 2x2 pixels, 3x3 pixels, or other suitable super-pixel sizes. In the cropping mode **108b**, the digital zoom module **108** crops a portion of the overall image, such as a quarter, a third, half, or other suitable cropping sizes. In the interleaving mode **108c**, the digital zoom module **108** combines the techniques of the binning mode **108a** and the cropping mode **108c**. Depending on the implementation, the digital zoom module **108** may determine the appropriate binning, cropping, or interleaving based on the input image **450** resolution and a desired output image resolution. The output image resolution may be based on an appropriate resolution for barcode decoding. As such, the output image resolution may be 0.25 megapixels, 0.5 megapixels, 1 megapixel, 2 megapixels, 3 megapixels, 4 megapixels, or any other suitable image for barcode decoding. In a particular preferred embodiment, the output image resolution is between 0.5 megapixels and 2 megapixels.

Referring next to FIG. 10, a flowchart **1000** illustrates a method for controlling object imaging and range finding by controller **107**, which controls each of the autofocus module **220**, aiming module **170**, and illumination module **180**. For the sake of clarity, FIG. 10 is discussed with regard to the controller **107**, autofocus module **220**, aiming module **170**, and illumination module **180**. However, any similarly suitable controller, autofocus module, aiming module, and illumination module may be used.

At block **1002**, the imaging engine device **100** detects a presence of an aim light pattern in the FOV. In some implementations, the imaging engine device **100** detects the presence through communication between the controller **107** and the aiming module **170**. In further implementations, the controller **107** receives an indication of the presence from the imaging system **110**. Depending on the implementation, the imaging engine device **100** detects a presence of an aim light pattern (e.g., an aiming dot and/or a visual indication of a beam) emitted from the imaging engine device **100** and controlled by the aiming module **170**. After detecting the presence of the aim light pattern, the flow proceeds to block **1004**. At block **1004**, the controller **107** determines a parallax target distance of an object of interest using the position of the aim light pattern. In some implementations, the controller **107** determines the target distance based on the size of the aim light pattern in the FOV. In other

implementations, the controller **107** determines the target distance based on how bright the aim light pattern is. In still other implementations, the aim light pattern is a complex pattern on the target and the controller **107** determines the target distance based on the pattern. For example, the pattern may be a series of vertical lines that the controller **107** determines the target distance on using apparent distance between lines.

Depending on the implementation, the imaging engine device **100** may include or be communicatively coupled to a screen, such as mobile device screen, computer screen, or screen included as part of the housing of the imaging engine. In some such implementations, the controller **107** causes the screen to display the determined target distance to a user. As such, the imaging engine device **100** may function as a range finder. In some implementations, the controller **107** only causes the screen to display the determined target distance when a range finding mode is enabled by the user. In further implementations, the controller **107** may cause the assembly to emit an audio cue in addition to or in place of causing the screen to display the determined target distance. For example, the imaging engine device **100** may read the target distance aloud for the user or may emit different noises to indicate different ranges (i.e., for 0-5 inches, 5-20 inches, 20-50 inches, etc.).

Next, at block **1006**, the controller **107** causes a lens assembly of the imaging system **110** to focus a variable focus optical element on the object of interest. In some implementations in which the imaging system **110** includes a ball-bearing motor lens, the controller **107** sends an indication to the ball-bearing motor lens to focus on the object based on the target distance. Depending on the implementation, block **1006** may occur before, after, or substantially simultaneously with part of block **1004**.

At block **1008**, the controller **107** causes a digital zoom module **108** and/or the imaging system **110** of the imaging engine **100** to select and operate in a zoom operation mode based on the target distance. The selection and operation may be based on a determination by the controller **107** to begin or change zoom mode operation. In some implementations, the image taken by the imaging system **110** has a resolution greater than the preferred resolution for barcode decoding. For example, the image taken by the imaging system may have a resolution of 4 megapixels while a preferred resolution for barcode decoding is 1 megapixel. The zoom operation mode may be one mode out of multiple different zoom modes. In some implementations, the zoom operation modes include at least a Near FOV mode, a Far FOV mode, and an interleaved mode. Depending on the implementation, the zoom operation modes may correspond with the level of digital zoom. For example, the imaging system **110** may be fully zoomed out (i.e., no zoom) when operating in the Near FOV mode and may be fully zoomed in (i.e., zoomed in 2-3 times) when operating in the Far FOV mode.

While operating in the Near FOV mode, the controller **107** or the imaging engine **100** may operate by binning pixels in images taken by the imaging system **110**. In some implementations, the imaging engine **100** performs 2x2 binning, i.e., combining pixels in a 2 pixel by 2 pixel square into a single super-pixel. The imaging engine **100** may perform 3x3 binning, 4x4 binning, or any other suitable binning. In other implementations, the imaging engine **100** performs binning proportional to the factor of difference between the image reading resolution and the barcode decoding resolution (i.e., 2x2 binning is preferred for a resolution difference of 4 megapixels vs. 1 megapixel).

While operating in the Far FOV mode, the controller **107** or the imaging engine **100** may operate by cropping a portion of the image. In some implementations, the imaging engine **100** crops a smaller portion of the image depending on the distance of the object, up to one quarter of the total area of the image. In further implementations, similar to binning in the Near FOV mode, the imaging engine **100** performs cropping proportional to the factor of difference between the image reading resolution and the barcode decoding resolution (i.e., a minimum cropping size of one quarter is preferred for a resolution difference of 4 megapixels vs. 1 megapixel).

While operating in the interleaved mode, the controller **107** or the imaging engine **100** may operate by interleaving cropping a portion of the image and binning pixels as described above. In some implementations, the imaging engine **100** may crop up to a quarter of the image and may bin pixels up to a 2x2 binning process, depending on the resolution of images taken by the imaging system **110** and the preferred resolution for barcode decoding. Depending on the implementation, the cropping and the binning may be performed alternatively, simultaneously, or one after another.

The controller **107** determines what operation mode the imaging engine **100** is to operate in based on the determined target distance of the object of interest. In some implementations, the controller **107** compares the determined target distance of the object to one or more threshold values in determining which zoom operation mode a zoom module **108** and/or imaging system **110** of the imaging engine **100** is to operate in. For example, the controller **107** may cause the imaging engine **100** to operate in the Near FOV mode when the target distance is below a first threshold value, the Far FOV mode when the target distance is above a second threshold value, and the interleaved mode when the target distance is between the two threshold values. In some such implementations, the imaging engine operates in the Near FOV mode when the target distance is less than or equal to 8 inches, the Far FOV mode when the target distance is greater than or equal to 40 inches, and the interleaved mode when the target distance is between 8 and 40 inches, non-inclusive.

Similarly, at block **1010**, the controller **107** causes the imaging engine **100** to select and operate in an illumination operation mode based on the target distance. The illumination operation mode may be one mode out of multiple different illumination modes. In some implementations, the illumination operation modes include at least a Reduced Power mode, a Near Illumination mode, and a Far Illumination mode. In some such implementations, the illumination module **180** alternates between two illumination fields depending on the illumination operation mode. For example, the illumination module **180** may provide a first illumination field when operating in either of the Reduced Power or Near Illumination modes and may provide a second illumination field when operating in the Far Illumination mode. Similarly, the illumination module **180** may instead provide a first illumination field when operating in the Near Illumination mode and a second illumination field when operating in the Reduced Power or Far Illumination modes.

The controller **107** determines what operation mode the imaging engine **100** is to operate in based on the determined target distance of the object of interest. In some implementations, the controller **107** compares the determined target distance of the object to one or more threshold values in determining which illumination operation mode the imaging engine **100** is to operate in. For example, the controller **107**

may cause the imaging engine **100** to operate in the Reduced Power mode when the target distance is below a first threshold value, in the Far Illumination mode when the target distance is above a second threshold value, and in the Near Illumination mode when the target distance is between the first and second threshold values. In some such implementations, the imaging engine **100** operates in the Reduced Power mode when the target distance is less than or equal to 24 inches, the Far Illumination mode when the target distance is more than or equal to 40 inches, and the Near Illumination mode when the target distance is between 24 and 40 inches, non-inclusive.

In some implementations, the controller **107** may determine that an object switches between two zoom and/or illumination operation modes. In such implementations, the controller **107** may change the operation mode correspondingly. In some such implementations, rather than immediately switch between illumination modes, the controller **107** instead implements a delay period. As such, when the controller **107** determines that the imaging engine **100** should change illumination operation modes (i.e., determines that the target distance surpasses or falls below a threshold value), the controller **107** waits for a predetermined period before switching modes. Depending on the implementation, the delay period may be 0.1 seconds, 0.5 seconds, 1 second, 5 seconds or any other similarly suitable length of time. In further implementations, the delay period resets each time the controller **107** determines that the imaging engine **100** should change illumination operation modes. For example, a target may be located approximately the threshold distance away between two modes. A user operating the imaging engine may move the reader back and forth, causing the controller **107** to read target distances in both operation mode brackets before settling in one. After the delay period passes fully in the final operation mode bracket, the controller **107** changes the illumination operation mode.

Though blocks **1008** and **1010** are described in one order, each of blocks **1008** and **1010** may occur substantially simultaneously or in any order between themselves. Similarly, in some implementations, blocks **1008** and **1010** may occur substantially simultaneously with, before, or after block **1006**.

After determining the operation mode or modes in which the imaging engine device **100** should operate, the controller **107** may cause one or more elements of the imaging engine device **100** to capture an image including the target in the FOV. For example, the controller **107** may cause the aiming module **170** to direct an aim light pattern onto a target before causing the autofocus module **220** to focus on the target. The controller **107** may then cause the illumination module **180** to operate in an illumination mode before causing the digital zoom module **108** and/or the imaging system **110** to zoom on the target and capturing an image. In implementations in which the target is a barcode, QR code, or other similar coded image, then after the imaging engine device **100** captures the image and/or crops an ROI of the image, the controller **107** decodes the target.

The above-identified imaging engine device **100** can be implemented in the barcode reader of FIGS. **11** and **12**. FIGS. **11** and **12** are exemplary embodiments of an optical imaging reader **500** (also referred to as a barcode reader) and the components thereof. However, it will be understood that the above-identified imaging engine is not exclusively implemented in barcode readers **500**, and is instead able to be implemented in any such device employing an image assembly with a field-of-view (FOV). With more specific

reference to barcode readers, it will be further understood that, although a particular embodiment of a barcode reader **500** is disclosed, this disclosure is applicable to a variety of barcode readers, including, but not limited to, gun-type handheld readers, mobile computer-type readers, presentation readers, etc.

Referring now to the drawings, FIG. **11** illustrates an exemplary barcode reader **500** having a housing **502** with a handle portion **504**, also referred to as a handle **504**, and a head portion **506** (also referred to as a scanning head **506**). Depending on the implementation, the housing **502** includes or is the chassis **150**. The head portion **506** includes a window **508**, and is configured to be positioned on the top of the handle portion **504**. In some implementations, the window **508** may be the window **266** and/or the window **310** discussed with regard to FIGS. **3** and **7** above. The handle portion **504** is configured to be gripped by a reader user (not shown) and includes a trigger **510** for activation by the user. Optionally included in an embodiment is a base (not shown), also referred to as a base portion, which may be attached to the handle portion **504** opposite the head portion **506**, and is configured to stand on a surface and support the housing **502** in a generally upright position. The barcode reader **500** can be used in a hands-free mode as a stationary workstation when it is placed on a countertop or other workstation surface. The barcode reader **500** can also be used in a handheld mode when it is picked up off the countertop or base station, and held in an operator's hand. In the hands-free mode, products can be slid, swiped past, or presented to the window **508** for the reader to initiate barcode reading operations. In the handheld mode, the barcode reader **500** can be moved towards a barcode on a product, and the trigger **510** can be manually depressed to initiate imaging of the barcode.

Other implementations may provide only handheld or only hands-free configurations. In the embodiment of FIG. **11**, the reader **500** is ergonomically configured for a user's hand as a gun-shaped housing **502**, though other configurations may be utilized as understood by those of ordinary skill in the art. As shown, the lower handle **504** extends below and rearwardly away from the body **502** along a centroidal axis obliquely angled relative to a central FOV axis of a FOV of an imaging assembly within the scanning head **502**.

For at least some of the reader embodiments, an imaging assembly includes a light-detecting sensor or imager **511** operatively coupled to, or mounted on, a printed circuit board (PCB) **514** in the reader **500** as shown in FIG. **12**. Depending on the implementation, the imaging assembly may be or may include imaging system **110**. Similarly the PCB **514** may be the PCB or circuit board **102** of the scan engine device **100**. In an embodiment, the imager **511** is a solid-state device, for example, a CCD or a CMOS imager, having a one-dimensional array of addressable image sensors or pixels arranged in a single row, or a two-dimensional array of addressable image sensors or pixels arranged in mutually orthogonal rows and columns, and operative for detecting return light captured by an imaging lens assembly **515** over a field of view along an imaging axis **517** through the window **508**. In some implementations, the imaging lens assembly **515** includes elements of or the entirety of the rolling shutter rolling shutter sensor system **300**. Similarly, in some implementations, the imager **511** is the imager **325** and/or an imager of imaging system **110**. The return light is scattered and/or reflected from a target **513** over the field of view. The imaging lens assembly **515** is operative for focusing the return light onto the array of image sensors to

enable the target **513** to be read. In particular, the light that impinges on the pixels is sensed and the output of those pixels produce image data that is associated with the environment that appears within the FOV (which can include the target **513**). This image data is typically processed by a controller (usually by being sent to a decoder) which identifies and decodes decodable indicial captured in the image data. Once the decode is performed successfully, the reader can signal a successful “read” of the target **513** (e.g., a barcode). The target **513** may be located anywhere in a working range of distances between a close-in working distance (WD1) and a far-out working distance (WD2). In an embodiment, WD1 is about one-half inch from the window **508**, and WD2 is about thirty inches from the window **508**.

An illuminating light assembly may also be mounted in the imaging reader **500**. The illuminating light assembly includes an illumination light source, such as at least one light emitting diode (LED) **519** and at least one illumination lens **521**, and preferably a plurality of illumination LEDs and illumination lenses, configured to generate a substantially uniform distributed illumination pattern of illumination light on and along the target **513** to be read by image capture. In a preferred embodiment, the illuminating light assembly is illumination module **180**, described in detail with regard to optical assembly **400** of the illumination module **180** in FIG. **8** above. At least part of the scattered and/or reflected return light is derived from the illumination pattern of light on and along the target **513**.

An aiming light assembly may also be mounted in the imaging reader **500** and preferably includes an aiming light source **523**, e.g., one or more aiming LEDs or laser light sources, and an aiming lens **525** for generating and directing a visible aiming light beam away from the reader **500** onto the target **513** in the direction of the FOV of the imager **511**. In a preferred embodiment, the aiming light assembly is aiming module **170** as described with regard to FIG. **6** above.

Further, the imager **511**, the illumination source **519**, and the aiming source **523** are operatively connected to a controller or programmed microprocessor **107** operative for controlling the operation of these components. A memory **529** is connected and accessible to the controller **107**. Preferably, the microprocessor **107** is the same as the one used for processing the captured return light from the illuminated target **513** to obtain data related to the target **513**. Though not shown, additional optical elements, such as collimators, lenses, apertures, compartment walls, etc. As discussed with regard to FIGS. **1-9** are provided in the head portion **506** of the housing. Although FIG. **12** shows the imager **511**, the illumination source **519**, and the aiming source **523** as being mounted on the same PCB **514**, it should be understood that different embodiments of the reader **500** may have these components each on a separate PCB, or in different combinations on separate PCBs. For example, in an embodiment of the reader, the illumination LED source is provided as an off-axis illumination (i.e., has a central illumination axis that is not parallel to the central FOV axis).

The above description refers potential embodiments of the accompanying drawings. Alternative implementations of the examples represented by the drawings include one or more additional or alternative elements, processes and/or devices. Additionally or alternatively, one or more of the example blocks of the diagrams may be combined, divided, re-arranged or omitted. Components represented by the blocks of the diagrams are implemented by hardware, software, firmware, and/or any combination of hardware, software and/or firmware. In some examples, at least one of the

components represented by the blocks is implemented by a logic circuit. As used herein, the term “logic circuit” is expressly defined as a physical device including at least one hardware component configured (e.g., via operation in accordance with a predetermined configuration and/or via execution of stored machine-readable instructions) to control one or more machines and/or perform operations of one or more machines. Examples of a logic circuit include one or more processors, one or more coprocessors, one or more microprocessors, one or more controllers, one or more digital signal processors (DSPs), one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), one or more microcontroller units (MCUs), one or more hardware accelerators, one or more special-purpose computer chips, and one or more system-on-a-chip (SoC) devices. Some example logic circuits, such as ASICs or FPGAs, are specifically configured hardware for performing operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits are hardware that executes machine-readable instructions to perform operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits include a combination of specifically configured hardware and hardware that executes machine-readable instructions. The above description refers to various operations described herein and flowcharts that may be appended hereto to illustrate the flow of those operations. Any such flowcharts are representative of example methods disclosed herein. In some examples, the methods represented by the flowcharts implement the apparatus represented by the block diagrams. Alternative implementations of example methods disclosed herein may include additional or alternative operations. Further, operations of alternative implementations of the methods disclosed herein may combined, divided, re-arranged or omitted. In some examples, the operations described herein are implemented by machine-readable instructions (e.g., software and/or firmware) stored on a medium (e.g., a tangible machine-readable medium) for execution by one or more logic circuits (e.g., processor(s)). In some examples, the operations described herein are implemented by one or more configurations of one or more specifically designed logic circuits (e.g., ASIC(s)). In some examples the operations described herein are implemented by a combination of specifically designed logic circuit(s) and machine-readable instructions stored on a medium (e.g., a tangible machine-readable medium) for execution by logic circuit(s).

As used herein, each of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium” and “machine-readable storage device” is expressly defined as a storage medium (e.g., a platter of a hard disk drive, a digital versatile disc, a compact disc, flash memory, read-only memory, random-access memory, etc.) on which machine-readable instructions (e.g., program code in the form of, for example, software and/or firmware) are stored for any suitable duration of time (e.g., permanently, for an extended period of time (e.g., while a program associated with the machine-readable instructions is executing), and/or a short period of time (e.g., while the machine-readable instructions are cached and/or during a buffering process)). Further, as used herein, each of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium” and “machine-readable storage device” is expressly defined to exclude propagating signals. That is, as used in any claim of this patent, none of the terms “tangible

machine-readable medium,” “non-transitory machine-readable medium,” and “machine-readable storage device” can be read to be implemented by a propagating signal.

In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive, and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

Moreover in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a,” “has . . . a,” “includes . . . a,” “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially,” “essentially,” “approximately,” “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each

claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

The invention claimed is:

1. A method for an imaging engine having an imaging assembly having a field of view (FOV), the method comprising:

detecting, by a microprocessor, a presence of an aim light pattern within the FOV;

determining, by the microprocessor and in response to the detecting, a target distance of an object in the FOV based on a position of the aim light pattern in the FOV, the target distance being a distance from the imaging engine to the object;

causing, by the microprocessor, a variable focus optical element to focus on the object based on the target distance; and

responsive to making a first determination, by the microprocessor, selecting, based on the target distance, one of a plurality of zoom operation modes,

wherein the plurality of zoom operation modes includes at least two of: (i) an image binning mode, (ii) an image cropping mode, and (iii) an image interleaving mode.

2. The method of claim 1, wherein selecting one of the plurality of zoom operation modes includes:

responsive to determining the target distance to be less than a lower threshold value, selecting the image binning mode;

responsive to determining the target distance to be greater than an upper threshold value, selecting the image cropping mode; and

responsive to determining the target distance to be between the lower threshold value and the upper threshold value, selecting the image interleaving mode.

3. The method of claim 2, wherein the lower threshold value is at most 12 inches and the upper threshold value is at least 24 inches.

4. The method of claim 1, wherein the variable focus optical element is one of a ball-bearing motor lens, a liquid lens, a T-len, or a voice coil motor lens.

5. The method of claim 1, wherein the object is a barcode, further comprising:

cropping a region of interest (ROI) including the barcode; and

decoding the barcode.

6. The method of claim 1, further comprising displaying, to a user, the target distance on a display communicatively coupled to the microprocessor.

7. An imaging engine having an imaging assembly having a field of view (FOV) and comprising:

a variable focus optical element disposed along an optical axis to receive light from an object;

an imaging sensor disposed along the optical axis to receive light from the variable focus optical element;

a digital zoom module configured to modify an image received from the imaging sensor;

an aiming module configured to generate and direct an aim light pattern; and

a microprocessor and computer-readable media storing machine readable instructions that, when executed, cause the imaging engine to:

detect a presence of the aim light pattern in the FOV; responsive to the detecting, determine a target distance of the object in the FOV based on a position of the

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aim light pattern in the FOV, the target distance being a distance from the imaging engine to the object; and
 responsive to making a first determination, select, based on the target distance, one of a plurality of zoom operation modes;
 wherein the variable focus optical element, the digital zoom module, and the aiming module are communicatively coupled to the microprocessor,
 wherein selecting one of the plurality of zoom operation modes includes:
 responsive to determining the target distance to be less than a lower threshold, selecting an image binning mode;
 responsive to determining the target distance to be greater than an upper threshold value, selecting an image cropping mode; and
 responsive to determining the target distance to be between the lower threshold value and the upper threshold value, selecting an image interleaving mode.
 8. The imaging engine of claim 7, wherein the digital zoom module is configured to, responsive to selecting the image binning mode, bin pixels of the image using at least one of: 2×2 pixel binning, 3×3 pixel binning, or 4×4 pixel binning.
 9. The imaging engine of claim 7, wherein the digital zoom module is configured to, responsive to selecting the image cropping mode, crop a portion of the image sized to at least one quarter of the image.
 10. The imaging engine of claim 7, wherein the digital zoom module receives the image with a resolution of at least 3 megapixels and zooms on the image with a resolution in a range of 0.5 to 2 megapixels.
 11. The imaging engine of claim 7, wherein selecting the zoom operation mode includes:
 responsive to determining the target distance to be less than a first lower threshold, selecting an image binning mode;
 responsive to determining the target distance to be greater than a first upper threshold value, selecting an image cropping mode; and
 responsive to determining the target distance to be between the first lower threshold value and the first upper threshold value, selecting an image interleaving mode.
 12. The imaging engine of claim 11, wherein the first upper threshold value and the second upper threshold value are equal.

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13. The imaging engine of claim 12, wherein the first upper threshold value and the second upper threshold value are at least 40 inches, the first lower threshold value is at most 8 inches, and the second lower threshold value is at most 24 inches.

14. The imaging engine of claim 7, wherein the imaging sensor is a rolling shutter sensor configured to operate in at least (i) a first state wherein an obfuscator of the rolling shutter sensor obfuscates a majority of radiation propagating along the optical axis and (ii) a second state wherein the obfuscator of the rolling shutter sensor transmits a majority of radiation propagating along the optical axis.

15. The imaging engine of claim 14, wherein the rolling shutter sensor is communicatively coupled to the microprocessor, and wherein the machine readable instructions, when executed, further cause the imaging engine to transition the rolling shutter sensor between the first state and the second state.

16. The imaging engine of claim 14, wherein the rolling shutter sensor has a pixel size of at most 2.0 micrometers.

17. The imaging engine of claim 7, wherein the aiming module includes at least:

a beam source assembly having a beam source for generating the aim light pattern from an exit surface, wherein the exit surface defines a central axis along which an input light is to propagate; and

a collimator assembly having a lens group that defines a tilt axis, wherein the tilt axis has a tilt angle relative to the central axis and the lens group is positioned to deflect the aim light pattern from the central axis onto the tilt axis.

18. The imaging engine of claim 7, wherein the aiming module generates and directs the aim light pattern in a pulsed laser driving mode.

19. The imaging engine of claim 7, wherein the aim light pattern has a wavelength of at least 505 nanometers and at most 535 nanometers.

20. The imaging engine of claim 7, wherein the variable focus optical element is one of a ball-bearing motor lens, a liquid lens, a T-len, or a voice coil motor lens.

21. The imaging engine of claim 20, wherein the variable focus optical element is the ball-bearing motor lens having a pupil diameter of at least 2.0 millimeters and a focus range from 3 inches to infinity.

22. The imaging engine of claim 7, wherein the object is a barcode and wherein the machine readable instructions, when executed, further cause the imaging engine to decode the barcode.

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